AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT131
Discrete Mathematics

Final Examination
Spring Semester 2024/2025

Date: 16/June/2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **50**
- This exam consists of **5** questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Q–1: [5×2 marks] Choose the correct answer:

1.1: $\neg (P \leftrightarrow Q)$ is logically equivalent to

a) $Q \leftrightarrow P$
b) $P \leftrightarrow \neg Q$
c) $\neg P \leftrightarrow \neg Q$
d) $\neg Q \leftrightarrow \neg P$
e) None of the above

1.2: The statement "x is greater than or equal to 2, for each real number x" can be represented by:

a) $\forall x \in \mathbb{R}, x \geq 2$
b) $\exists x \in \mathbb{R}, x \geq 2$
c) $\forall x \in \mathbb{R}, x \leq 2$
d) $\neg \forall x \in \mathbb{R}, x \leq 2$
e) None of the above

1.3: The quotient and remainder when -5 is divided by 6 is

a) $q = 5, r = 6$
b) $q = 0, r = 6$
c) $q = 0, r = 5$
d) $q = -1, r = 1$
e) None of the above

1.4: Which of the following is an equivalent to $(U \cap A) \cap (\emptyset \cup \bar{A})$

a) U
b) A
c) B
d) $\emptyset$
e) None of the above

1.5: The number of edges in the complete bipartite graph $K_{3,5}$ is

a) 5
b) 10
c) 15
d) 25
e) None of the above

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Each question is worth 10 marks. Answer the following questions:

Q–2:

a) **[5 marks]** Construct the truth table of the statement:

$$\neg(p \rightarrow q) \rightarrow \neg q$$

Determine whether the statement is a tautology.

b) **[3x1 marks]** Determine the truth value of each of the following statements:

i. If 5 is a rational number then $5 \div 2 = 5$.

ii. $\forall x\,(x + 2 > x)$, domain is the set of integers.

iii. $18 \equiv 7\ (mod\ 5)$

c) **[2x1 marks]** Let $U = \{1,2,3,4,5,6,7,8,9,10\}$, $A = \{4,5,7,8,9\}$, and $B = \{3,5,9,10\}$. List the element of the following:

i. $\bar{A} \cup \bar{B}$

ii. $\bar{\phi}$

Q–3:

a) **[2+2+1 marks]** Suppose that a "password" is any string of 5 letters from the (26) alphabet, with repeated letters allowed.

i. How many passwords are there?

ii. How many passwords end with A?

iii. How many passwords begin with B or C and end with B or C?

b) **[2+2+1 marks]** Suppose that two integers from the set $\{1,2,\ldots,6\}$ are chosen at random. Find the probability that:

i. 2 and 6 are picked.

ii. Both numbers match.

iii. Sum of the two numbers picked is less than 4.

Q–4:

a) **[2+3 marks]** Let R be a binary relation on $A = \{1,2,3,4\}$ given by the matrix:

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

i. Sketch the directed graph of R.

ii. Is R equivalence relation? Justify your answer.

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b) **[2+2+1 marks]** Consider the poset $(\{2, 3, 4, 6, 10, 12, 16\}, |)$.

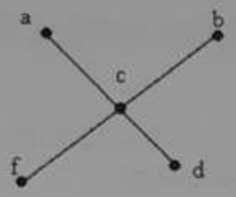
i. Draw the Hasse diagram.

ii. What is the upper bound of $\{2, 4\}$?

iii. What is the lower bound of $\{2, 4\}$?

Q–5:

a) **[3+3 marks]** Given a graph $G = (V, E)$ with vertices a, b, c, d, f :



i. Write the adjacency matrix of G.

ii. Write the incident matrix of G.

b) **[4 marks]** Set up a binary search tree to sort the following numbers:
22, 8, 14, 17,3,9,27,11

End of questions

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